MSc. in Computing
Practicum Approval Form

Section 1: Student Details

Project Title:	Text-to-Symbolic Conversion for Legal Statutory Reasoning
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Chosen major:	Natural Language Processing
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Date of Submission	24 November, 2024

Section 2: About your Practicum

Please answer all questions below. Please pay special attention to the word counts in all cases.

What is the topic of your proposed practicum? (100 words)

The problem we are addressing is the limited access to legal services due to their high cost and complexity. Large Language Models(LLMs) hold great potential to improve access to justice. However, a major challenge in applying AI and LLMs in legal contexts, where consistency and reliability are crucial, is the need for System to reasoning. Our research will focus on improving statutory reasoning in AI systems using a neuro-symbolic approach. We aim to enhance the accuracy, scalability, and accessibility of legal information. This includes leveraging LLMs to generate high-quality logical representations from legal texts and exploring hybrid methods that combine LLMs with logic programming.

Please provide details of the papers you have read on this topic (details of 5 papers expected).

1. **"How Does NLP Benefit Legal System: A Summary of Legal Artificial Intelligence"**: This paper discusses the transformative potential of NLP in the legal domain, including automating document review and legal research, predicting case outcomes, and summarising law documents. It highlights datasets like Supreme Court databases and legislative records, which are used to train ML models for legal applications.

2. **“Connecting Symbolic Statutory Reasoning with Legal Information Extraction”**: The paper addresses the challenge of connecting statutory reasoning with automated legal information extraction by translating natural language case descriptions into structured Prolog knowledge bases (KBs). It proposes an improved ontology for annotating legal cases and adapts state-of-the-art information extraction (IE) models to extract structured representations from the SARA dataset. The study finds that enhanced IE performance correlates with improved accuracy in statutory reasoning tasks, demonstrating the utility of structured representations.
3. **“Reliable Reasoning Beyond Natural Language”**: The paper proposes a neuro-symbolic approach to improve reasoning in large language models (LLMs) by integrating Prolog, a logic programming language, into their inference pipeline. This method translates problem statements into Prolog code, enabling explicit deductive reasoning that addresses the shortcomings of LLMs’ sequential text generation. It demonstrates improved performance on benchmarks like GSM8k and introduces the Non-Linear Reasoning (NLR) dataset, which evaluates LLMs on complex reasoning tasks requiring logical and contextual understanding. The approach highlights how combining symbolic and neural reasoning enhances reliability and scalability in tasks where LLMs traditionally struggle.
4. **“Can GPT-3 Perform Statutory Reasoning?”:** The paper investigates GPT-3’s ability to perform statutory reasoning using the SARA dataset and synthetic statutes. While GPT-3 achieves better results than prior methods on SARA, it still makes significant errors, particularly with simple synthetic statutes it has not encountered before. The study highlights GPT-3’s limitations in understanding unseen legal rules and its struggles with reasoning tasks requiring precise logical application, suggesting a need for further research to improve LLMs’ capabilities in statutory reasoning.
5. **“Equitable Access to Justice- Logical LLMs Show Promise”:** The paper explores how large language models (LLMs) integrated with logic programming can improve access to justice by making legal reasoning more interpretable and precise. It highlights the challenges in encoding legal contracts into logical representations and compares the performance of two OpenAI models, showing that the newer model significantly outperforms its predecessor in translating legal texts into Prolog rules. The study demonstrates how such neuro-symbolic AI approaches can help scale computable legal frameworks, offering potential solutions to address barriers to legal access.

How does your proposal relate to existing work on this topic described in these papers? (200 words)

Our proposal builds on existing work to address challenges in statutory reasoning with large language models (LLMs), using the SARA (Statutory Reasoning Assessment) dataset evaluates LLMs’ ability to reason over legal rules and queries. While LLMs upgrading improves general performance, issues with logical consistency and interpretability remain.

To enhance performance, we propose a neuro-symbolic approach that integrates LLMs with logic programming (e.g., Prolog). This method uses LLMs to generate logical representations of legal rules, enabling precise and interpretable reasoning through symbolic systems. By combining the strengths of both neural and symbolic methods, our approach aims to improve accuracy on SARA while addressing scalability and real-world applicability in legal contexts.

What are the research questions that you will attempt to answer? (200 words)

Our research focuses on integrating natural language processing (NLP) and computational law to automate legal queries and enhance statutory reasoning. We aim to address two key questions:

- How can logic-based languages like Prolog be effectively utilized to encode legal statutes for automated reasoning? This includes assessing the ability of Prolog to represent legal rules and its integration with NLP systems for query translation and reasoning.
- To what extent can the neuro-symbolic approach improve an AI system's statutory reasoning capabilities? We will investigate how combining large language models (LLMs) with symbolic reasoning methods enhances accuracy, consistency, and interpretability in handling legal rules and queries.

How will you explore these questions? (Please address the following points. Note that three or four sentences on each will suffice.)

- What software and programming environment will you use?

We will use Python as the primary programming language, leveraging state-of-the-art LLMs(e.g., GPT-4, GPT-4o, Claude Sonnet 3.5) to reproduce experiments and generate symbolic representations of legal texts. Frameworks like TensorFlow and PyTorch will be used for deep learning tasks, while tools such as SQLAlchemy and Prolog engines (e.g., SWI-Prolog) will support database and symbolic reasoning tasks. Overleaf will be utilized for collaborative documentation, GitHub for version control, and Zotero with the Google Scholar connector for managing references and bibliographies in BibTeX format.

- What coding/development will you do?

Rule-based logic will be developed to enable parsing and reasoning for symbolic tasks. Additionally, we will design workflows to integrate symbolic reasoning systems with neural networks, ensuring accurate and scalable legal reasoning capabilities. This includes creating pipelines for generating logical encodings of legal texts and testing them against benchmarks like SARA.

- What data will be used for your investigations?

The SARA (Statutory Reasoning Assessment) dataset is a benchmark designed to evaluate AI systems' ability to perform statutory reasoning. It consists of 9 statutory sections drawn from the U.S. tax code(i.e. title 26 of the U.S. Code) and 376 hand-crafted cases that state simple facts, involving taxpayers named Alice, Bob, Charlie, and Dan, and ask one question that can be answered by applying some of the nine sections to the facts.

SARA also includes a translation of legal statutes, factual scenarios, and corresponding questions to Prolog, requiring models to interpret and apply legal rules to specific cases. SARA tests reasoning capabilities such as rule application, logical consistency, and handling exceptions, making it a critical resource for developing and assessing AI solutions in the legal domain.Solving the questions via Prolog results in 100% accuracy.

- Is this data currently available, if not, where will it come from?

Yes, the SARA dataset is publicly available and has been extensively used in research on statutory reasoning and legal benchmarks.

- What experiments do you expect to run?

We plan to conduct two sets of experiments to evaluate and compare the performance of current AI models and our proposed neuro-symbolic approach for statutory reasoning:

1. **Baseline Experiments:** Recreating Experiments from Blair-Stanek et al.:
    - Dataset: Run experiments on both the SARA dataset and the synthetic dataset created in their study.
    - Methodology:
        - Use GPT-4, GPT-4o, or Claude Sonnet 3.5 in place of the models used in the original experiments.
        - Reuse the code available in the linked GitHub repository to replicate the process, modifying only the models.
        - Evaluate the models’ performance on tasks such as applying legal rules and reasoning through exceptions.
    - Objective: Establish a baseline for statutory reasoning tasks using state-of-the-art LLMs.
    - Metrics: Accuracy, logical consistency, and ability to handle unseen statutes.
2. **Neuro-Symbolic Approach Experiments**
    - Dataset: Applying Neuro-Symbolic Approach to SARA Dataset
    - Methodology:
        - Provide the LLM with a relevant statute and have it generate a Prolog knowledge base (Prolog code).
        - Provide the LLM with a legal question and have it convert it into a Prolog query.
        - Use a Prolog solver to answer the query and retrieve results.
    - Comparison:
        - Assess the LLM’s ability to generate correct Prolog code for statutes and queries.
        - Compare reasoning accuracy and consistency with the baseline models from the first experiment.
    - Objective: Evaluate the effectiveness of combining symbolic reasoning with LLMs in improving performance on statutory reasoning tasks.

These experiments will help benchmark state-of-the-art LLMs against a neuro-symbolic framework to determine whether integrating symbolic reasoning improves accuracy, interpretability, and scalability in statutory reasoning tasks. if it does, to what extent it will be.

- What output do you expect to gather?

- Insights into whether the neuro-symbolic approach improves reasoning accuracy, scalability, and interpretability.
- Clear benchmarks for comparing neural and neuro-symbolic approaches in statutory reasoning tasks.
- Recommendations for future model improvements based on observed strengths and limitations.

- How will the results be evaluated?

The results will be evaluated using quantitative metrics such as accuracy, precision, recall, logical consistency, and execution success rates for tasks on the SARA and synthetic datasets. The quality of Prolog knowledge bases and queries generated by the models will be analyzed for syntactic and semantic correctness, alongside error analysis to identify limitations. A comparative evaluation will assess the performance of baseline models (GPT-4, GPT-4o, Claude Sonnet 3.5) against the neuro-symbolic approach, focusing on accuracy, scalability, and interpretability. Additionally, real-world applicability will be considered by examining the alignment of outputs with legal reasoning requirements, ensuring the system’s utility in handling complex statutes and queries.